

1.6 Classwork: Sampling bias and survey methodologies

Answer in IB terminology. The sampling techniques are: simple random, convenience, systematic, quota and stratified. Data are either discrete or continuous.

1. Alex is writing a mathematical exploration (IA) on the sleep habits of the students at a high school. The number of students in each grade is shown in the table.

Grade	9	10	11	12	Total
Number of students	84	78	68	70	300

- (a) Write down the population for Alex's investigation.
- (b) Alex first plans to survey the 30 students in the Grade 12 common room at lunch. State the name of this sampling technique and explain why it is likely to produce a biased sample.
- (c) Alex decides instead to take a stratified sample of 30 students by grade.
- (i) Show that 8 students will be selected from Grade 10.
- (ii) Find the number of students selected from each of the other grades.
- (d) To choose the Grade 12 students, Alex lists their names in alphabetical order, picks a random starting name among the first ten, and then selects every 10th name on the list. State the name of this sampling technique.

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Nickname:

2. Sam is collecting survey data for an exploration on whether screen time is related to hours of sleep. For each student Sam records the number of apps on their phone and the time, in hours, they slept last night.
- (a) State whether each variable is discrete or continuous.
 - (i) The number of apps on a phone.
 - (ii) The time slept last night.
 - (b) Sam stands at the school entrance and surveys the first 15 boys and the first 15 girls who arrive. State the name of this sampling technique.
 - (c) Explain why the technique in part (b) is not the same as a stratified sample by gender.
 - (d) The school office gives Sam a numbered list of all 300 students. Describe how Sam could use this list to select a simple random sample of 30 students.
 - (e) One student records a sleep time of 19 hours. Suggest what Sam should do with this data value, and give a reason.
 - (f) One survey question reads: “Don’t you agree that using your phone late at night ruins your sleep?” Explain why this question may produce biased data.

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